

1. For each of the following, determine whether the set is open, whether it is closed, and whether it is compact. You don't have to prove your answers.
 - (a) $\mathbb{Z}$
 - (b) $\mathbb{Q}$
 - (c) $\mathbb{R}$
 - (d) $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} \cup \{0\}$
 - (e) $(0, 1) \cup [3, 4]$
 - (f) $\{2023\}$
2. Suppose that $\{O_\alpha\}_{\alpha \in A}$ is a collection of open sets, and $\{C_\alpha\}_{\alpha \in A}$ is a collection of closed sets.
 - (a) Prove that $\cup_{\alpha \in A} O_\alpha$ is open.
 - (b) Is it true that $\cup_{\alpha \in A} C_\alpha$ is closed? Prove or provide a counterexample.

3. For sets A and B , define $A + B = \{a + b : a \in A, b \in B\}$.
- (a) Prove that if A and B are open, then $A + B$ is open.
 - (b) Prove that if A and B are compact, then $A + B$ is compact.
 - (c) It is *not* true that the sum of closed sets must be closed. Provide a counterexample.

4. A set $E \subset \mathbb{R}$ is said to be *monochromatic* if every continuous function $f : E \rightarrow \{-1, 1\}$ is constant.
- (a) Show that $[0, 1]$ is monochromatic.
 - (b) Give an example of a set that is not monochromatic.